BioSynvia System Requirements Document

Version 0.2

November 5, 2026. Working title. Owner: Lakshmi Sindhu Pulugundla.

System objective

Implement the offline CHO soft-sensor workbench in PRD v1.2 and a separately gated package of reusable bioprocess modeling components. The component package combines typed process data, model execution, numerical verification, registry and versioned research outputs. Physical-system synchronization, closed-loop control and operational Digital Twin validation remain outside scope.

Edition control and requirement status

Part A is the controlling v0.2 expansion. Part B reproduces SRD v0.1 in full, retaining every SR-DAT, SR-FEA, SR-MOD, SR-UI, SR-MH and SR-NFR ID and acceptance test T-001 through T-018. Original version/page labels identify baseline content. PRD v1.2 controls product scope. This SRD defines proposed system behavior and tests; none of the new SR-DT requirements is declared implemented.

Baseline location	Effective v0.2 change
Sections 1 through 3	Add Track B component architecture and release distinction
Sections 4.1, 4.2 and 6	Add raw-assay missingness invariants, unit-operation and modality metadata, explicit availability semantics
Sections 4.5, 7 and 8	Add callable simulation, evidence tiers, registry/consumer, characterization and bounded optimization
Sections 9 and 10	Retain T-001 through T-018; add T-019 through T-034 and new DT traceability
Sections 11 through 13	Add CG1 through CG5 and component-release acceptance; retain Track A gates

Current implementation evidence

Archive verification, source filename reconciliation, the provisional audit CLI and project-owned EDA script exist. EDA found 214 original assay blanks represented as aligned zeros and incomplete temporal sensor coverage. The 18 current regression tests are software evidence only, not completion of T-001 through T-018. Target approval, canonical/model-ready release, model fitting, component simulation and dashboard acceptance remain open. [E1]

Track B B0 requirements are required for the component release; B1 requirements need acceptance or an approved deferral. Gated requirements must emit an eligibility decision. Synthetic component verification cannot be counted as real-data model validation.

1 Component architecture and canonical extensions

The offline flow is source verification, raw-aware data audit, canonical records, approved target/features, group-safe evaluation, component registration, bounded execution, validated outputs and independent consumer/read-only review. No module may bypass source, target, split or evidence-tier checks by calling the numerical solver directly through the supported interface.

Logical entities and invariants

Entity	Required information	Invariant
UnitOperation	ID, type, material context, scale/context metadata and provenance	Unknown material/scale attributes explicit; no invented CGT identity
Measurement	Experiment/reactor/unit-operation IDs, field/variable IDs, inline/at_line/offline modality, event and availability time, values/units/quality/source refs	Availability unknown remains null with reviewed assumption; never autoequal to event time
ParameterSet	ID/version, component version, named values and units, fit groups, bounds, covariance/uncertainty where available, approval	Only compatible immutable versions; synthetic and fitted values distinguishable
Component	ID/version/family, state/input/output schemas, equations artifact hash, applicability domain and evidence tier	No dashboard dependency; no untrusted executable loading
Scenario	ID/version, component/parameter refs, initial state, input schedule, times, objective/constraints, seed and budget	Immutable request plus source/assumption provenance
ResearchArtifact	Schema/type/version, payload hashes, parents, status, units, timestamps and claim limits	No measured/simulated label ambiguity; missing outputs never fabricated

Added data controls

SR-DT-001 (B0; DT-01). The system shall validate these entities and referential links, including unit-operation/material context and the acquisition-versus-availability distinction. Missing required references or incompatible units shall block execution. Unknown metadata may be explicit only where the schema permits it. Evidence: T-019 and T-020.

SR-DT-002 (B0; DT-01 and DT-06). Canonical assay validity shall derive from original sheets. A blank original assay shall retain unknown_missing_cause, null canonical value and source pointer even if an aligned cell is zero. A zero present in the original record shall remain a distinct observed value, subject to quality review. Record original and aligned values separately. Evidence: T-021.

SR-DT-003 (B0; DT-01 and DT-03). Feature eligibility shall combine acquisition/availability policy, original sensor quality status, accepted alignment policy, required-channel support and reviewed maximum age. Stale/missing/ineligible features shall produce unsupported status or a preregistered missing-feature policy. Do not treat any prior sensor row as proof of feature validity. Evidence: T-020 and existing T-005/T-010.

2 Reusable execution and numerical requirements

Model interface

SR-DT-004 (B0 and G7; DT-02). A public local interface shall expose describe, validate and simulate operations independent of the dashboard. A request binds the component version, initial state, process-input schedule, parameter version, output-time grid and solver settings. A consumer shall invoke it without duplicating equations. Invalid requests fail before numerical execution. Evidence: T-022.

SR-DT-005 (B0 and G7; DT-02). Each mechanistic component shall define equations, state meaning/units, parameter units and signs, feed/volume/dilution conventions and applicability bounds. Numerical verification shall include a known analytic limiting case, tolerance refinement, state/flow consistency and declared conservation/positivity rules. If a conservation law is intentionally approximate, document the omitted terms and accepted diagnostic. Scientific error budgets shall be justified before acceptance, not invented after fitting. Evidence: T-023 and T-024.

SR-DT-006 (B0; DT-02 and DT-03). Eligibility shall resolve to synthetic_verified, data_calibrated, group_evaluated or unsupported, with source and review evidence. G7 failure shall block data-backed calibration claims while permitting an isolated synthetic demonstration. Hybrid outputs inherit limitations of their mechanistic and correction components. Evidence: T-025.

Comparative evaluation

SR-DT-007 (B0 and G7; DT-03). Statistical, mechanistic and hybrid comparisons shall share a frozen group manifest, observed-target mask, target unit and declared information scenario. Model-family differences in initialization or assay inputs shall create separate comparison strata. Preprocessing, calibration, correction and parameter selection use training groups only. Unsupported families are not assigned fabricated errors. Evidence: T-026 and T-007/T-008/T-009.

SR-DT-008 (B0; DT-03 and DT-06). Evaluation shall retain per-experiment errors, coverage/abstention counts, macro group summaries, fit failures, eligibility outcomes and negative baseline deltas. Uncertainty distinguishes numerical, parameter, predictive and scenario uncertainty. An absent interval shall be null with a reason, never an implied zero-width interval. Evidence: T-026 and existing T-011/T-012.

Failure behavior

Solver failure, physically inadmissible states, unsupported bounds, unknown units or insufficient data shall return explicit status and reason codes. Partial solver output may be retained as unapproved diagnostics, but no approved prediction/trajectory shall be emitted for the failed request. All failures retain request IDs, parameter/model refs and reproducible diagnostics.

3 Registry outputs and downstream consumer

SR-DT-009 (B0; DT-04). Registry metadata shall bind model/component, equations/code hashes, parameter set, feature/target contracts, training groups, environment, schema versions, evidence tier, domain and approval records. Any changed dependency invalidates approval/cached results. No entry marked approved_for_research implies manufacturing validation. Evidence: T-027.

SR-DT-010 (B0; DT-04). Metadata shall be JSON validated against published versioned schemas; tabular trajectory/prediction payloads shall be typed Parquet with declared units/time semantics and finite-or-null values. Serialize measured, predicted and simulated origins separately. Unsupported point outputs must be null, not zero or NaN. Evidence: T-019, T-027 and T-028.

SR-DT-011 (B0; DT-04). A separate local consumer shall validate schemas and payload digests and read outputs without importing private scientific implementation modules. Accept supported schema versions only; reject unknown major versions, digest mismatches, missing required units and unresolved parameter refs. Safe loading rules in SR-NFR-011 remain in force. Evidence: T-028.

Proposed simulation request contract

Required top-level keys: schema_version, request_id, unit_operation_id, component_ref, parameter_set_ref, evidence_mode, initial_state, input_schedule, output_times_hours, solver and provenance_refs. Named state/input/parameter values are paired with units; reference objects carry ID, version and SHA-256. No example below is an observed dataset value.

Output times shall be finite and increasing within the permitted horizon. Input schedules shall define interpolation/hold semantics and event order, cover the requested horizon and reject overlapping ambiguous segments. Units must resolve through the approved conversion registry. Evidence: T-019, T-022 and T-023.

Proposed result envelope

```
{
  "schema_version": "1.0.0",
  "artifact_kind": "simulation",
  "request_id": "synthetic-contract-example",
  "execution_status": "unsupported",
  "evidence_tier": "unsupported",
  "trajectory_ref": null,
  "predicted_value": null,
  "uncertainty": null,
  "reason_codes": ["PARAMETER_SET_NOT_APPROVED"],
  "research_only": true
}
```

Successful envelopes additionally require component/parameter/schema refs, units, time origin, applicability, numerical diagnostics, source/parent hashes, creation time, environment and uncertainty-status metadata. Null fields must satisfy the status-specific schema; unsupported examples do not exempt successful outputs from those fields.

4 Characterization comparability and troubleshooting

SR-DT-012 (B0; DT-06). A characterization protocol shall define analysis windows, growth/trajectory summaries, units, valid data masks and experiment-level uncertainty. Summarize paired reactors within experiments; record process mode, design settings, quality events and confounders. Invalid growth-rate windows shall return unavailable with a reason. Evidence: T-029.

SR-DT-013 (B1 and G; DT-07). Comparability requests shall contain a defined change, attributes/units, group assignments, estimand, margins with scientific rationale, uncertainty/multiplicity plan, assay support and reviewer. Missing prerequisites force descriptive_only or insufficient_evidence status. A nonsignificant difference must never automatically yield equivalent. VCD differences shall not be labeled product-quality comparability without the required quality attributes. Evidence: T-030.

SR-DT-014 (B0; DT-06). Troubleshooting outputs shall separate observations, candidate hypotheses and required follow-up. Status flags, long gaps and all-zero channels shall retain source evidence and not become an automated causal diagnosis. Evidence: T-029.

Characterization artifacts

Produce characterization_protocol.json, per_experiment_characterization.parquet, sensor_quality_summary.parquet and process_characterization_report.md. Each summary carries group IDs, time windows, valid counts, units, quality/exclusion rules and source/config hashes. Compare mode-associated patterns descriptively unless adjusted estimates have adequate independent-group support.

The current 12-group design does not justify treating 274,781 sensor rows as independent replication. Growth-phase selection, transformations and exclusion decisions must be logged. Restrict any bootstrap to a justified experiment/paired structure and state the limitations of inference with three fed-batch groups.

Comparability artifacts

Store comparison_protocol.json and comparability_report.md with a status enum: descriptive_only, insufficient_evidence, inconclusive or protocol_criterion_met. The last means only that a prespecified research criterion was met. It must not be rendered as regulatory approval or product comparability without separate appropriate evidence and review. Reviewer-defined margins are required; no default biological threshold is supplied.

Current EDA issue disposition

The 214 raw blanks/aligned zeros require SR-DT-002 regression fixtures. Fourteen post-recording assays and 27 assays with prior sensor age over one hour motivate SR-DT-003; one hour is a diagnostic sensitivity level, not an approved universal maximum age. Entirely zero fitted C channels require quality review, not silent deletion. These findings are descriptive evidence from E1, not accepted scientific thresholds.

5 Sensitivity scenarios and constrained optimization

SR-DT-015 (B1 and G; DT-05). Sensitivity, sweep and optimization requests shall specify model/parameter versions, evidence tier, initial states, decision variables/units, objective/units/direction, constraints, bounds/domain provenance, seed, solver and evaluation/time budget. Freeze the protocol before the run. Missing objective, undefined units or unapproved operating bounds block optimization. Evidence: T-031.

SR-DT-016 (B1 and G; DT-05). Validate schedules, domain, numerical state limits and constraints both before and after simulation. Return feasible, infeasible, unsupported, numerical_failure or budget_exhausted. A failed candidate may not be labeled optimal. Confirm returned feasible candidates with tighter solver tolerances and sensitivity to model/parameter uncertainty, and report boundary hits. Evidence: T-031 and T-032.

SR-DT-017 (B1; DT-05). Store every evaluated scenario, objective, feasibility result, failure and stopping reason in an immutable local ledger with complete lineage. Do not suppress nonconvergence or use held-out experimental outcomes to optimize settings. Report model-conditional optima as hypothetical and unconfirmed by experiment. Evidence: T-032.

Bounded execution policy

The scientific protocol must supply finite bounds and a finite evaluation/time budget. No automatic cloud compute, external service, scheduling or live process actuation is introduced. Existing no-cloud-cost rules remain. A local timeout or exhausted budget yields a terminal status and retained diagnostics, not a silently restarted unlimited search.

Positive-state checks, conservation checks and solver tolerances are component-specific. Small numerical negatives may be flagged within an approved tolerance; do not clip them silently. Include validation that the proposed solution is feasible under the accepted numerical policy and that claimed benefit does not disappear under reasonable tolerance refinement.

UI and service separation

The simulation executor is a local CLI/library. The read-only dashboard displays previously generated scenarios and comparisons with visible hypothetical labels. Dashboard launch must not fit, calibrate, optimize, mutate registry records or contact process equipment. This preserves SR-UI-007 and SR-NFR-009. A later interactive computation design requires a new explicit scope decision and read/write separation.

Model risk disclosure

Observed calibration fit, held-out prediction, numerical verification and optimization feasibility are different evidence types. Neither a converged solver nor a feasible optimizer establishes biological usefulness. If parameter identifiability is inadequate, block data-backed scenario recommendations and retain only clearly separated synthetic demonstrations.

6 Conditional scale PAT and downstream extensions

SR-DT-018 (G; DT-08). Produce scale_translation_feasibility.md identifying scales, relevant geometry/engineering variables, comparable biology, units and evidence gaps. No cross-scale validation status is permitted without a suitable dataset, independently defined scale groups and a transfer-evaluation protocol. Volume differences alone shall not create kLa, mixing, shear or power-input measurements. Evidence: T-033.

SR-DT-019 (G; DT-09). A spectroscopy adapter contract shall include modality, instrument/calibration metadata, spectral axis/intensity units, sample and reference IDs, measurement availability, experiment/batch groups and source hashes. Preserve a separate dataset namespace; absent spectroscopy data yields data_unavailable rather than simulated Raman validation. Evidence: T-034.

SR-DT-020 (G; DT-09). PCA/PLS or other chemometric preprocessing, component selection and reference alignment shall be train-group-only and point-in-time valid. Do not merge unrelated CHO and spectroscopy records as if they were paired observations. Missing units/reference linkage block a scientific modeling claim. Evidence: T-034 and existing T-007.

SR-DT-021 (Future G; DT-10). A downstream extension shall require a separate chromatography source, unit-operation/material contract, operating variables, chromatogram/analytical outputs, source rights, group design and evaluation protocol. Until approved, return not_implemented or data_unavailable and produce a documented extension contract/gap record. Evidence: T-034.

Extension evidence statuses

Each extension shall distinguish contract_defined, data_unavailable, synthetic_verified, data_calibrated, group_evaluated and unsupported. Data-gap reports are accepted feasibility deliverables; they do not count as executed scientific validation. Separate datasets retain separate provenance and evaluation scopes even when consumed by a common registry.

Artifacts and dependency references

Artifact	Producer	Required parents
component_contract.json	Component authoring	Schema version, equations/code ref, intended use
parameter_set.json	Calibration/review	Component, fit-group manifest, units and bounds
simulation_request.json	Scenario authoring	Component, parameter set, input/domain protocol
result.json and trajectory.parquet	Execution	Request, source refs, environment and solver
consumer_validation.json	Independent consumer	Result/schema/payload digests
scale/PAT/downstream feasibility reports	Extension audit	Source inventory, required fields and gap decisions

Paths are local and manifest-controlled. Use atomic approved-artifact promotion after complete validation; interrupted writes remain unapproved. Do not load arbitrary serialized executables to satisfy a registry reference.

7 Added acceptance tests and product traceability

T-001 through T-018 remain required where applicable. These new tests are proposed specifications; no execution pass is claimed here.

Test	Fixture and required result	Requirements
T-019	Missing unit-operation ref, incompatible unit or wrong schema type/version is rejected; valid typed contract passes	SR-DT-001/010; DT-01/04
T-020	A result collected before cutoff but available after cutoff is excluded; unknown availability needs reviewed policy	SR-DT-001/003; DT-01/03
T-021	Original blank plus aligned zero stays missing; original measured zero stays observed with provenance	SR-DT-002; DT-01/06
T-022	Separate caller invokes component without copying equations; invalid input never reaches solver	SR-DT-004; DT-02
T-023	Known analytic limiting case and unit-consistent balance fixture match approved numerical tolerances	SR-DT-005; DT-02
T-024	Tolerance refinement and state/conservation violation trigger diagnostics or rejection as specified	SR-DT-005; DT-02
T-025	Failed G7 or unidentifiable parameter set cannot become data-backed evidence; synthetic output stays labeled	SR-DT-006; DT-02/03
T-026	Family comparison with different groups, target masks or input scenarios is rejected or separated; negative result retained	SR-DT-007/008; DT-03
T-027	Mutated parameter/code/payload hash invalidates lineage; missing uncertainty is explicit	SR-DT-009/010; DT-04
T-028	Independent consumer reads valid outputs and rejects corrupt payloads/unsupported major schema versions	SR-DT-010/011; DT-04
T-029	Unequal reactor/group sizes preserve paired experiment summaries; inadequate growth window returns unavailable	SR-DT-012/014; DT-06
T-030	Missing comparison margins or quality attributes forces descriptive/insufficient status; nonsignificance does not imply equivalence	SR-DT-013; DT-07
T-031	Out-of-domain, incompatible-unit, unconstrained or infeasible scenario is rejected; failed point not labeled optimum	SR-DT-015/016; DT-05

Test	Fixture and required result	Requirements
T-032	Budget exhaustion returns terminal status; all evaluations retained; feasible candidate rechecked numerically	SR-DT-016/017; DT-05
T-033	Single-scale/volume-only data produces gap report and blocks cross-scale validation claim	SR-DT-018; DT-08
T-034	Missing spectra/references/chromatography data yields gap status; train/test or cross-dataset mixing fails	SR-DT-019/020/021; DT-09/10

Track B release coverage is all ten DT product IDs through SR-DT-001 through SR-DT-021. B0 engineering requirements must pass; scientific conditional branches require documented eligibility/no-go. A deferral is not a passing test. The QA report shall distinguish implemented, tested, blocked, failed and deferred requirements.

8 Release acceptance source register and baseline

Component release sequence

CG1 requires the raw-aware data/characterization pack and disposition of EDA issues. CG2 requires component contracts, equations/state/unit review and G7 evidence tier. CG3 requires numerical verification, versioned registry/output schemas and independent consumer tests. CG4 requires fair family comparisons where eligible plus bounded scenario/optimization evidence or approved B1 deferral. CG5 assembles the reproducible research component package, extension gap reports and claims review.

Retain the original source, target, split, holdout, naming and data-rights gates. Track A acceptance does not assert Track B acceptance. A complete component engineering package may contain a synthetic_verified simulation but shall not be labeled a validated CHO mechanistic/hybrid model. No target approval, independent scientific review or release sign-off is fabricated by this document update.

Required reviewer evidence package

Provide the full source/quality manifest, target and feature approvals, group protocol, component/equations contract, parameter/identifiability record, solver verification, applicable model comparisons, characterization report, comparability status, scenario ledger, registry and consumer results, pinned environment and exact reproduction commands. Include negative results, unsupported states and extension data gaps. Acceptance records identify reviewer, date, purpose, scope and artifacts.

Added sources and evidence

R9. NIST publication record for Satwekar et al., Operationalizing Digital Twins in Biomanufacturing through Interoperable Process Analytical Technology, July 16, 2026; DOI 10.1007/s00449-026-03369-9. Architectural context only; no standards-conformance claim.

https://www.nist.gov/publications/operationalizing-digital-twins-biomanufacturing-through-interoperable-process

R10. FDA PAT framework guidance, October 2004. Used to frame process measurements and modality interfaces, not as regulatory approval.

https://www.fda.gov/regulatory-information/search-fda-guidance-documents/pat-framework-innovative-pharmaceutical-development-manufacturing-and-quality-assurance

R11. FDA/ICH Q5E guidance, June 2005. Used to distinguish exploratory trajectory comparison from product comparability around manufacturing changes.

https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q5e-comparability-biotechnologicalbiological-products-subject-changes-their-manufacturing-process

E1. BioSynvia reports/eda-v2/report.md, summary.json, source_manifest.json, raw_aligned_assay_comparison.csv and scripts/eda.py. Supports descriptive local data findings; does not validate model predictions or exhaustively audit original inline status streams.

Part B retained SRD v0.1

The following 25 pages reproduce the complete baseline unchanged. Read baseline statements such as proposed not built and optional P2 together with this edition's current-status and two-track scope rules. The baseline's FR/SR/test IDs remain stable. References to PRD v1.1 in retained pages mean the retained foundation as controlled by PRD v1.2 Part A.

End of controlling revision. Retained baseline follows.

BioSynvia | System Requirements Document

Reproducible Bioprocess Soft-Sensing & Hybrid Modeling Workbench

Independent Portfolio Research Project
November 4, 2026

Problem statement and intended solution

The problem. Chinese hamster ovary (CHO) cell-culture experiments produce frequent bioreactor sensor measurements, but biologically important attributes may be available only at intermittent offline assay times. A model might estimate the missing state between assays, but apparent performance can be inflated by future-data leakage, data from related reactors appearing in both training and testing, or evaluation based on interpolated labels rather than actual assays.

System objective. BioSynvia (working title) shall provide a reproducible, offline research pipeline that determines whether available inline sensor observations can estimate a *verified, measured* biological target in **unseen CHO cultivation experiments**. It shall compare simple/statistical/ML methods under group-safe validation, disclose errors and uncertainties, and present results with transparent provenance through a read-only dashboard. A mechanistic/hybrid extension is conditional on experimental data suitability.

Expected result. A traceable scientific research prototype: source manifest, vetted target and units, canonical records, frozen experiment-group splits, reproducible baselines and candidate models, per-experiment metrics, failure analysis, model card, read-only user interface and independently repeatable demonstration. **It is not a manufacturing control application, a qualified soft sensor, a production Digital Twin, or a clinically validated product.**

Document control and interpretation

Field	Value
System / working title	**BioSynvia** — name not legally cleared for public commercial branding
Document	System Requirements Document (**SRD**); includes implementation-oriented software requirements
Document version	v0.1, November 4, 2026
Authoritative product source	**BioSynvia PRD v1.1 — Problem Statement First** (November 4, 2026)
System lifecycle status	**Proposed**, not built; all interfaces/schema examples are designs, not observed outputs
System owner / planned developer	Lakshmi Sindhu Pulugundla
Primary data source	Uhlendorff et al., CHO-K1 cultivation dataset, DOI 10.5281/zenodo.20829178 [R1]

Field	Value
Deployment	Local Python CLI + optionally local Streamlit web interface; no cloud dependencies in V1
Document authority	Derives implementation requirements from PRD; does **not** supersede unresolved PRD gates
Approval state	**DRAFT FOR SCIENTIFIC AND TECHNICAL REVIEW**

Interpretation legend: SOURCE = explicitly specified by PRD v1.1 or its cited dataset description. **PROPOSED DESIGN** = engineering detail recommended in this SRD. **GATED** = must not be treated as settled until audit, scientist review, data rights check, or other recorded decision. An SRD requirement is a plan, not a claim that BioSynvia currently meets it.

Traceability notation: PRD FR-01 denotes a functional requirement in PRD §6; PRD NFR-01 denotes a quality requirement in §10; G0–G9 are PRD §11 decision gates. SRD identifiers remain stable across versions even if their wording changes.

1. Purpose, scope and system boundaries

1.1 Purpose and intended readers

This SRD translates BioSynvia's scientific/product objectives into implementable and testable system behavior. It is intended for the developer, scientific reviewer, QA reviewer, and a hiring manager evaluating scientific rigor. It establishes required artifacts, workflows, interfaces, invariants, acceptance tests and release restrictions; detailed implementation choices remain subject to controlled technical-design decisions.

1.2 In scope for the first presentable research version (V1)

- Offline ingestion of a legally obtained CHO source archive with provenance and integrity verification.
- Experiment/reactor inventory; original/aligned time, units, missingness and data-quality audits.
- Recorded selection of a real offline assay target based on coverage and scientific meaning.
- A typed canonical representation of experiments, paired reactors, sensors and assays.
- **Sensor-only inference:** at time t, use only validated inline data available at or before t.
- Experiment-group partitioning, versioned/frozen manifests and training-only preprocessing.
- Simple baselines, Ridge and/or PLS, and one supported nonlinear candidate; group-aware model comparison.
- Per-experiment model performance, missingness/abstention, negative findings and limitations.
- Versioned artifact and evidence manifests, tests, locally reproducible commands and read-only dashboard.

1.3 Conditional advanced extension (V1.1/P2)

Mechanistic ODEs, identifiable parameter estimation, mechanistic–statistical hybrid correction and hypothetical scenarios are out of V1's required scope. They become data-calibrated claims only after PRD gate **G7/MH-01** confirms observed states, correct units, appropriate feeds/volumes and identifiability. Otherwise they may be explored **only** as explicitly synthetic, educational simulations.

1.4 Out of scope and prohibited interpretations

No live manufacturing connection; no automatic control action; no patient/clinical decision; no release-quality, GMP/GxP or FDA-compliant validation claim; no real-time/continuous production telemetry; no commercial product

or open-data redistribution before rights review; no self-declared Cell and Gene Therapy work experience; no model performance claim before the model has run. The project is **CHO**, not CAR-T/AAV, and no affiliation with Johnson & Johnson or another company may be implied.

1.5 Operational assumptions and unresolved questions

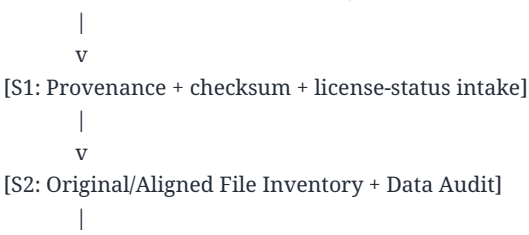
- **SOURCE:** the publisher describes 24 CHO-K1 cultivations from 12 paired experiments (9 batch, 3 fed-batch), 38 inline and 10 offline variables, and original Excel plus aligned CSV data [R1].
- **NOT YET VERIFIED BY IMPLEMENTATION:** precise CSV header names, observation units, target assay name, source license, feature availability, viable cell density coverage, timestamps, measured limits, device calibration and realistic prediction latency. These must be resolved through Phase P1.
- **GATED:** the 2-experiment final holdout is only a candidate protocol; if the data do not permit a defensible holdout, grouped leave-one-experiment-out with nested tuning must be reviewed and documented instead.
- **GATED:** business/scientific utility error thresholds cannot be invented. A scientist must select any target-specific acceptance threshold after unit and use-case review, before a utility claim.

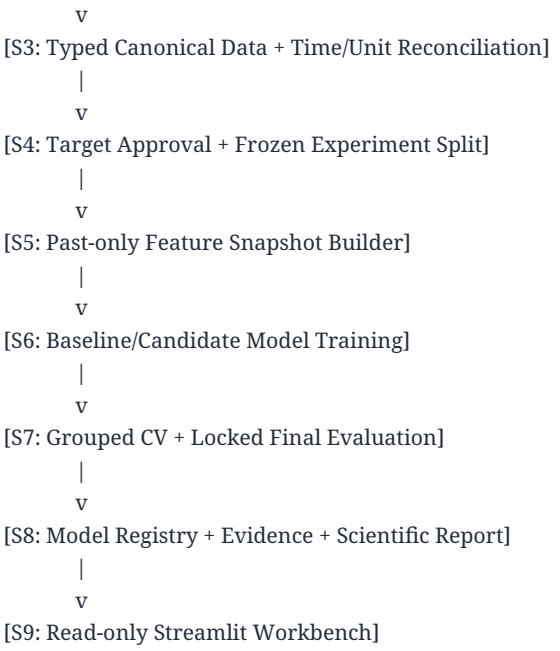
2. Stakeholders, actors and high-level use cases

Actor	Privilege in V1	Core use case	System response
Project developer / modeler	Local CLI; writes derived artifacts	Import, audit, train, evaluate and release approved report	Deterministic outputs with run manifest, tests and errors
Scientific reviewer	Reads reports and dashboards	Examine assay meaning, model assumptions and failure cases	Visible units, experiment groups, measured/estimated indicators and limitations
Data/automation reviewer	Reads schemas and provenance	Trace a prediction to its raw source, feature snapshot and model	Resolve hashes, schema versions and input timestamps
Hiring-manager reviewer	Read-only demo/report	Understand methodology and results	Honest, bounded narrative with negative results shown
Trademark/data-rights decision owner	Authorizes public release	Verify permissions and branding	Publication blocked until recorded clearances or neutral alternative

Use cases: UC-01 obtain/audit dataset; UC-02 choose target/inference contract; UC-03 construct leakage-safe feature snapshots; UC-04 freeze experiment splits; UC-05 train baselines/candidates; UC-06 compare and sign off scientific evaluation; UC-07 inspect predictions and failure cases; UC-08 reproduce an approved report; UC-09 optionally run mechanistic/hybrid research after approval.

3. Logical architecture and component responsibilities





Optional after G7: [S10: Mechanistic/Hybrid Research Module]

Component ID	Responsibility	Inputs	Outputs	Forbidden behavior
S1 Provenance	Verify archive checksum and track rights status	Local archive path, citation/config	source_manifest.json, hash audit	Implicitly mark unclear dataset rights as cleared
S2 Inventory/Audit	Reconcile source files, formats and experiment IDs	Raw archive/Excel/CSV	Inventory, column dictionary, quality report	Silently drop malformed/missing files
S3 Canonical store	Normalize field types, units and timestamps	Audited source files	Local canonical Parquet/CSV and schema manifest	Modify original source; invent units
S4 Research protocol	Approve target and freeze grouped partitions	Metadata + target decision	Target contract and immutable split manifest	Choose split using final test model outcomes
S5 Features	Build point-in-time sensor-only records	Canonical records + cutoff policy	Feature snapshots, lineage and rejection log	Read sensor values or assay targets from after cutoff
S6 Training	Fit baselines and candidates only on development groups	Features + fold plan + config	Versioned fitted pipelines and development metrics	Fit transformers globally or consult final test
S7 Validation	Compute identical metrics on identical groups	Model candidate + frozen test	Per-group errors, failures, aggregate report	Hide failed experiments or rebrand tuned holdout
S8 Provenance/reporting	Resolve results to reproducible evidence	Runs, configs, hashes	Model cards, reports, artifact manifest	Report simulated quantities as measured

Component ID	Responsibility	Inputs	Outputs	Forbidden behavior
S9 Dashboard	Inspect approved artifacts and qualify predictions	Approved local manifests, plots	Read-only figures/tables/warnings	Retrain silently, alter raw data, make control recommendations
S10 Mechanistic	Optional state/process models and sensitivity analyses	Verified states/units or synthetic fixtures	ODE/hybrid appendix and limitations	Claim real-process model when state evidence is insufficient

Architecture decision (PROPOSED): use a Python package (src/biosynvia), local filesystem artifacts (CSV/Parquet/JSON/YAML), scikit-learn/SciPy, pytest, and Streamlit. This stack is not a requirement to deploy cloud infrastructure or a database. A SQLite experiment index may be considered later, but the source of truth in V1 should remain immutable manifests and reproducible files.

4. System functional requirements — detailed and atomic

Priority definitions: P0 = necessary for a credible V1; P1 = important V1 robustness/traceability (must be addressed or expressly deferred with rationale); P2 = optional extension behind an explicit gate. **Status for every requirement below: PROPOSED / NOT IMPLEMENTED.**

4.1 Source integrity and dataset audit

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
SR-DAT-001	P0 · FR-01	System **shall** require a local archive and expected checksum before parsing source observations.	Valid checksum passes; corrupted byte or wrong digest fails with nonzero exit and no model-ready data.
SR-DAT-002	P0 · FR-01	System shall write DOI, authors/source citation, retrieved filename, expected and observed digest, timestamp, rights status and tool version to source_manifest.json.	Manifest validates against SourceManifest contract; missing rights status explicitly unknown.
SR-DAT-003	P0 · FR-02	System shall enumerate source Excel workbooks and aligned CSVs and reconcile expected experiment/reactor/modalities against discovered files.	Explicit report for 12 experiments × 2 reactors × 2 aligned modalities (=48 aligned CSVs) or documented dataset discrepancy; duplicate IDs fail validation.
SR-DAT-004	P0 · FR-03	System shall collect and version raw column names, inferred types, units where documented, source mapping and missingness for every candidate field.	Every selected field has a dictionary row and observed file references; unknown units remain unverified, never guessed.
SR-DAT-005	P0 · FR-04	System shall preserve source time variables and record the	Alignment report identifies resampling/offset

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
		original-to-aligned timestamp transformation or sampling policy for each file.	assumptions; mismatch exceeding approved tolerance is flagged.
SR-DAT-006	P0 · FR-03	System shall detect duplicate timestamps, invalid numeric values, unparseable dates and suspicious gaps without silently altering raw values.	Fixture containing each anomaly yields a record-level rejection/quality event and reconciled counts.
SR-DAT-007	P0 · FR-05	System shall block modeling until a versioned target decision records target source column, unit, coverage, assay meaning, sampling cadence and approval.	Training refuses missing/unsigned target config; report shows valid assay counts per experiment.
SR-DAT-008	P1 · NFR-03	System shall make ingestion idempotent for identical input archive/config and write derived artifacts atomically.	Re-run yields matching contents or versioned identical manifest; interrupted write leaves no approved partial output.

4.2 Canonical records and point-in-time features

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
SR-FEA-001	P0 · FR-03	Canonical observations shall contain experiment_id, reactor_id, measurement kind, original source field, normalized variable ID, value, units, event time and source reference.	Schema validation and referential integrity pass for all included records.
SR-FEA-002	P0 · FR-04	The target-alignment operation shall generate candidate scoring rows only at **actually measured offline assay timestamps** (and capture availability time where known).	Count of scored target rows equals accepted observed assays; interpolated target points never appear as ground truth.
SR-FEA-003	P0 · FR-07	A sensor-only feature snapshot at cutoff t shall read only eligible inline values with acquisition/availability time no later than t.	Perturb all future sensor values; all features and earlier predictions remain bitwise/equivalently unchanged.
SR-FEA-004	P0 · FR-07	Rolling-window aggregations (mean, median, slope, variability) shall be computed from past-only windows with explicit window, minimum	Unit fixtures compare exact expected windows; too-sparse windows return missing indicators or abstention, not invented samples.

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
		support and missingness rules.	
SR-FEA-005	P0 · FR-07	Feature eligibility shall be recorded for each input: source, unit, cutoff availability, transformation and leakage risk.	Versioned feature_eligibility.csv is complete for selected features; ineligible field cannot enter fitting.
SR-FEA-006	P0 · FR-06	Experiment-group partitioning shall use experiment_id as the indivisible grouping key, not reactor_id, point or file.	No experiment ID or its paired reactors overlap train/development/test in any fold.
SR-FEA-007	P0 · FR-09	A frozen split manifest shall contain experiment IDs, partition roles, mode labels, random seed where relevant, decision rationale and approval status.	Rerun selects same groups; failed approval blocks final test execution.
SR-FEA-008	P1 · FR-13	When critical inputs are missing, invalid or outside supported training ranges, prediction paths shall emit unsupported or abstained with an explicit reason.	Missing-sensor/out-of-range fixtures generate visible status and no fabricated numeric output.

4.3 Model fitting, comparison and evaluation

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
SR-MOD-001	P0 · FR-08	Implement a sensor-only naive baseline and at least one transparent single-sensor or simple regression baseline on the same eligible scoring rows.	Baseline scores saved with target units, sample counts and identical fold IDs.
SR-MOD-002	P0 · FR-08	Fit at least one linear/chemometric model (Ridge and/or PLS) and one feasible nonlinear model (Random Forest or gradient boosting).	Configurable pipeline fit and predicted values captured for the same grouped folds.
SR-MOD-003	P0 · FR-08/09	All imputers, scalers, selectors and hyperparameter searches shall fit **inside development training groups** only.	Spy transformer records fit group IDs; no validation/final test groups fitted.
SR-MOD-004	P0 · FR-09	Model-selection runs shall use frozen group-safe CV folds and preserve parameter grids, seed, software versions, code revision and raw/feature data	A run can be reproduced from an artifact manifest and named config.

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
		hashes.	
SR-MOD-005	P0 · FR-10	Evaluation shall compute per-experiment MAE, RMSE, signed bias, valid/abstained counts and baseline deltas in original target units.	Metrics agree with deterministic test fixture; group count reconciles with split.
SR-MOD-006	P0 · FR-10	Primary ranking shall use macro-average **experiment-level MAE**, not a pooled point-level error, and report per-experiment spread.	Unequal-size fixture demonstrates equally weighted experiments.
SR-MOD-007	P0 · FR-09	The reserved final holdout shall be evaluated only after the champion/config are approved, and shall record the first holdout evaluation immutably.	Unauthorized early holdout access fails; post-hoc retuning is labeled exploratory.
SR-MOD-008	P1 · FR-14	Evaluation shall report batch/fed-batch diagnostics, at least one sensor-family ablation and a missing-sensor stress test where scientifically meaningful.	Mode/sample counts and degraded/error results included, even when unfavorable.
SR-MOD-009	P1 · FR-13	If prediction intervals are produced, report empirical group-aware coverage, interval width and calibration assumptions.	Intervals absent or explicitly marked uncalibrated if insufficient calibration groups.
SR-MOD-010	P1 · FR-15	Each approved model artifact shall link model config, preprocessing pipeline, feature contract, training groups, environment lock and evaluation report.	Provenance lookup resolves complete chain; missing dependent artifact blocks approval.
SR-MOD-011	P0 · FR-10	The final report shall preserve unfavorable/negative outcomes and avoid declaring the soft sensor useful solely from favorable pooled metrics.	Report fixture includes model worse than baseline on a group; corresponding failure remains visible.

4.4 Read-only workbench and research communication

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
SR-UI-001	P0 · FR-11	Overview shall identify source dataset, approved target/unit, model version, experiment	Reviewer locates each within 60 seconds.

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
		count and research-only status.	
SR-UI-002	P0 · FR-11	Data Explorer shall filter experiment/reactor and distinguish measured assays, inline signals, aligned/interpolated inputs and missing intervals.	Markers/legends and explanatory text distinguish types without relying on color alone.
SR-UI-003	P0 · FR-11	Model Comparison shall separate development CV from untouched final test and show baseline and candidate per-group errors.	Final scores never appear under development label; failed groups remain visible.
SR-UI-004	P0 · FR-11/13	Prediction Review shall render measured/predicted series, residuals and unavailable/unsupported warnings; never plot fake ground truth.	Null target fixture renders missing state with explanation, not a smooth invented line.
SR-UI-005	P0 · FR-12	Every view presenting results shall display or prominently link research-only, non-validated, no-process-control limitations.	Manual UI checklist confirms disclaimer and status in all model-result views.
SR-UI-006	P1 · FR-15	Quality & Provenance view shall display dataset digest, split/model IDs, source/feature contracts, generated timestamp and model card.	Reviewer can trace an arbitrary shown prediction to relevant evidence in ≤5 minutes.
SR-UI-007	P0 · PRD §8	V1 dashboard shall be **read-only** against raw, canonical, feature, model and report artifacts.	UI action inspection and filesystem sandbox test prove no background retraining or mutation.
SR-UI-008	P1 · PRD §8	Plots and tables shall have textual labels, units, accessible status messages and no color-only distinctions.	Manual accessibility review and visual regression pass.

4.5 Conditional mechanistic and hybrid requirements

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
SR-MH-001	P2 · FR-16/G7	Before fitting to measured data, a signed mechanistic eligibility review shall confirm observed states, documented units, applicable assumptions,	Without eligible review the run mode is synthetic_only; cannot emit data-calibrated label.

SRD ID	Priority / PRD reference	Shall requirement	Verification and acceptance evidence
		sufficient experiments and identifiability diagnostics.	
SR-MH-002	P2 · FR-16	ODE solver shall document equations, initial conditions, parameter ranges, solver tolerances and unit-consistent mass-balance assumptions.	Deterministic solver and sign/conservation tests pass under documented fixtures.
SR-MH-003	P2 · FR-16	Parameter fitting shall implement multi-start/sensitivity or equivalent identifiability checks and record unstable solutions.	Fit status distinguishes converged/unstable/non-identifiable; no blanket “validated” status.
SR-MH-004	P2 · FR-17	Hybrid correction shall use the same frozen group splits, inference contract and evaluation metrics as its statistical and mechanistic comparators.	Leakage regression and paired comparison pass; hybrid underperformance preserved.
SR-MH-005	P2 · FR-18	Any scenario plot shall be labeled hypothetical and restricted to declared input bounds; no optimized manufacturing recipe recommendation.	Out-of-bounds parameters rejected; disclaimer visible in scenario export.

5. Non-functional requirements and operational quality

Requirements in this section inherit PRD §10. They are **quality specifications**, not claims about implementation. A P0 quality failure blocks a research-ready V1, even if the dashboard renders correctly.

SRD ID	Parent	Priority	System-level requirement	Objective verification
SR-NFR-001	NFR-01	P0	Given the same verified raw archive, approved configuration and pinned environment, the complete local evaluation shall reproduce the same partition IDs, scored observation counts and numerically equivalent metrics.	Two clean-environment runs agree on hashes of deterministic intermediates and metric values within documented floating-point tolerances.
SR-NFR-002	NFR-02	P0	Every reportable metric and displayed prediction shall resolve to source digest, target	Automated lineage traversal resolves every reference without a missing artifact.

SRD ID	Parent	Priority	System-level requirement	Objective verification
			decision, field dictionary, feature version, split ID, fitted pipeline ID, code revision and environment information.	
SR-NFR-003	NFR-03	P0	Original data shall remain byte-for-byte immutable and shall never be saved into a repository commit by default. Intermediate output shall be generated in separate directories with atomic writes.	Before/after source digests match; source/archive paths are excluded by repository rules.
SR-NFR-004	NFR-04	P0	No training or evaluation path shall cross an experiment-group boundary, access future sensor/target values, or score interpolated offline targets as measured truth.	Leakage injection tests fail unauthorized access, paired-reactor split tests fail overlap, and assay counts reconcile.
SR-NFR-005	NFR-05	P0	Invalid times, unit conflicts, missing required IDs, unsupported model schema versions and unavailable required sensors shall cause a typed error or explicit abstention; the system shall not silently coerce them.	Negative test fixtures produce named error codes and visible logs.
SR-NFR-006	NFR-06	P0	Failed model runs, inferior baselines, excluded experiments and unsupported predictions shall be retained and distinguishable from success cases.	Report contains failed-run ledger and negative comparison fixtures.
SR-NFR-007	NFR-07	P0	Raw and derived data shall be processed locally by default, with no patient data, unapproved telemetry	Static dependency and configuration audit finds no outbound telemetry or secrets; network access is not

SRD ID	Parent	Priority	System-level requirement	Objective verification
			or credential export.	required for analysis after download.
SR-NFR-008	NFR-08	P1	Results, tables and plots shall show units, definitions, accessible text and visual distinctions not relying exclusively on color.	Manual keyboard/contrast/label review; image descriptions or chart tables present.
SR-NFR-009	NFR-09	P0	The V1 execution path shall not invoke billable cloud services, scheduled remote automation or GitHub Actions without separate recorded owner authorization.	No active automatic workflow in supplied template; local CLI tests run offline.
SR-NFR-010	NFR-10	P0	Public-release tooling shall check naming review status and raw-data license status, and refuse to package source data if rights remain unknown.	Public package simulation excludes archive; branding approval field remains unverified.
SR-NFR-011	NFR-11	P0	Model loading shall accept only known local artifacts associated with a validated manifest and trusted project paths. Dependencies shall be inventoried with pinned version constraints.	Path traversal, corrupted artifact and untrusted serialized object fixtures are rejected.
SR-NFR-012	PRD §8	P1	Read-only dashboard startup on a reference local laptop shall reach a usable overview without retraining. Any performance target shall be measured and documented, not declared before implementation.	Cold-start measurement and no-write-filesystem test included in release report.

Risk-specific decision: Performance numbers such as “dashboard in under 2 seconds” or scientific tolerances such as “MAE below 5%” are intentionally **not** hardcoded before the dataset audit and target feasibility review. Setting arbitrary thresholds now would reward superficial test compliance instead of useful biological modeling.

6. Proposed data contracts and persistence model

6.1 Logical entities, keys and cardinalities

Entity	Required identifiers	Required metadata	Cardinality / invariant
Experiment	experiment_id	process_mode, source_id, optional design details	One per source experiment; exactly two reactor instances expected unless source audit proves a discrepancy
ReactorRun	experiment_id, reactor_id	Time origin, start/end, source file ID	Unique composite key; reactor never maps to multiple experiment IDs
Measurement	measurement_id, experiment_id, reactor_id	Variable ID, kind, source timestamp, elapsed hours, value, unit, quality flag, provenance	Multiple per reactor; primary uniqueness strategy documented; raw duplicate events not silently overwritten
OfflineAssay	assay_id, run key	Target variable, collected time, available time (if known), measured value/unit, assay status	Only valid actual assays can become ground-truth scoring rows
FeatureSnapshot	snapshot_id, run key, cutoff	Feature contract version, sensor values, window support, missingness, lineage	All constituent sensor observation availability times <= cutoff
SplitAssignment	split_version, experiment_id	Role, mode, seed/method, approval	Each experiment appears in exactly one partition per manifest
ModelRun	run_id, model_id	Target/version, environment, code/config hashes, feature and split refs, status	Run cannot be approved without complete dependency references
PredictionRecord	prediction_id, model_id, snapshot_id	Target/unit, point estimate, optional interval, support/abstention state	Invalid inference emits status and reason instead of numeric estimate
MetricRecord	run_id, experiment_id, metric_name	Value, unit, valid count, abstained count, reference baseline	All rows tied to actual observed assays and frozen group assignments

Source preservation policy. The upstream archive, every original Excel sheet and aligned CSV must retain a source-level identifier and digest. Converted Parquet tables are derived products. Their schema, timezone/cutoff assumptions, units and lineage must remain discoverable independently of a running model.

6.2 Machine-readable proposed schemas

The following JSON is a *design example*, not a claim that these values were observed in the CHO files. All field labels are placeholders pending the Phase 1 inventory.

```
{
  "schema_version": "1.0.0",
  "source_id": "cho_zenodo_20829178",
  "doi": "10.5281/zenodo.20829178",
```

```
 "archive_file": "Dataset.zip",
 "hash_algorithm": "md5",
 "expected_hash": "71b2a35b7df2ed11644ec9f177453772",
 "observed_hash": null,
 "retrieved_at_utc": null,
 "license_status": "unverified",
 "public_raw_data_distribution_allowed": false
}
```


A measurement object is expected to include experiment_id, reactor_id, source_field, variable_id, measurement_kind (inline or offline), observed_elapsed_hours, source_time_original, availability_elapsed_hours (nullable when unknown), value, unit_original, unit_normalized (nullable until reviewed), quality_status, and source_file_digest.

A **feature snapshot** contains snapshot_id, experiment/reactor IDs, cutoff_elapsed_hours, features as a mapping of typed named values, support_by_feature (count, age and missing status), source_sensor_refs and feature_contract_hash. A feature can never be computed with a right-inclusive window extending beyond its cutoff.

A **prediction envelope** must make invalidity explicit:


```
{
 "schema_version": "1.0.0",
 "experiment_id": "EXP_PLACEHOLDER",
 "reactor_id": "REACTOR_PLACEHOLDER",
 "cutoff_elapsed_hours": 120.0,
 "model_version": "model-placeholder",
 "prediction_status": "unsupported",
 "target_variable": "TO_BE_APPROVED",
 "target_unit": "TO_BE_APPROVED",
 "predicted_value": null,
 "prediction_interval": null,
 "reason_codes": ["TARGET_NOT_APPROVED"],
 "research_only": true
}
```


The example is intentionally an abstention. No made-up assay values or impressive-looking model scores are provided in the specification.

6.3 File paths and artifact manifests

The proposed root is a local folder outside public version control for raw data. Every generated artifact should have a stable, content-addressed or versioned identifier. A manifest must list file name, SHA-256 digest, schema version, producer tool version, parent artifact IDs and creation time.

Location	Contents	Publish?
data/raw/	Publisher ZIP and original files	**No** without rights confirmation
data/interim/	Parsed source records and alignment audits	No by default; may contain derived copies of source data
data/processed/	Modeling tables and snapshots	Only with cleared rights and explicit review

Location	Contents	Publish?
configs/	Target, split, feature and experiment configuration	Yes if credentials/source data are absent
runs/	Immutable local model runs, predictions and metrics	Selectively; verify licensing and no sensitive content
reports/	Data audit, evaluation, model card, release evidence	Yes after factual review
app/	Read-only dashboard source	Yes after naming review or under neutral title
tests/fixtures/	Synthetic minimal cases, including leak injection	Yes when generated without publisher data

6.4 Source identifier and unit policy

All source fields receive stable IDs from field_dictionary.csv, distinct from their display names. Units are never guessed from names such as X, VCD, CO2 or feed. Conversion requires a written transformation, conversion constant, dimensional compatibility review, and test fixture. A normalized value must preserve a pointer to its raw value and original unit. If a scientifically required variable's unit is unknown, any data-calibrated mechanistic component using it is blocked.

Missingness semantics: distinguish not_sampled, sensor_unavailable, parse_invalid, below_detection_limit, late_report, unit_unknown and excluded_by_protocol, to the extent the source supports them. If the source cannot distinguish these categories, retain unknown_missing_cause rather than inventing a reason.

7. Application interfaces and execution behavior

7.1 CLI contract (proposed, subject to implementation review)

Operation	Proposed command	Main inputs	Outputs	Failure behavior
Verify source	biosynvia verify-source --archive PATH --expected-md5 HEX	Raw ZIP, approved digest	Source manifest, checksum report	Nonzero exit on mismatch; no derived writes
Audit	biosynvia audit --archive PATH --output DIR	Verified source, mapping config	File inventory, field dictionary, quality reports	Missing/duplicate source IDs block approval
Choose target	biosynvia approve-target --config FILE	Approved scientific choice, reviewer record	Versioned target contract	No target -> train blocked
Build features	biosynvia features --target FILE --split FILE	Canonical tables, cutoff policy	Feature snapshots and eligibility log	Invalid time/unit or unauthorized field -> fail or abstain
Train	biosynvia train --config FILE	Frozen development folds, training config	Candidate pipelines, CV scores	Refuse any final-test access; preserve failed runs

Operation	Proposed command	Main inputs	Outputs	Failure behavior
Evaluate	biosynvia evaluate --model RUN_ID --holdout MANIFEST	Approved champion, locked test groups	Immutable final-test report	Holdout access requires approved gate; repeated tuning label blocked
Report	biosynvia report --run RUN_ID	Verified metric/lineage records	PDF/Markdown report and model card	Missing dependency -> cannot mark report approved
Launch UI	biosynvia dashboard --manifest FILE	Approved read-only artifacts	Local Streamlit process	Missing/corrupt model artifact shows warning; no retraining

A *dry-run* command should validate schemas and inspect dependencies without fitting models or consuming the holdout. Every CLI operation shall log a structured result with a named outcome (`success`, `blocked`, `invalid`, `abstained`, or `failed`) and a deterministic exit code.

CLI spellings are **design proposals** consistent with PRD §9.3; they do not imply installed commands exist.

7.2 API and user interface boundaries

The V1 dashboard reads a report manifest, tables, and precomputed approved predictions from local files. It must not call `fit()`, perform auto-tuning, modify raw data, rewrite artifacts, or calculate hidden new model outputs without making the computation/revision explicit. It should not collect or send viewer data to third-party analytics. Public HTTP access is not required and should remain disabled by default.

The UI information architecture comprises **Overview**, **Data Explorer**, **Model Comparison**, **Prediction Review**, and **Quality & Provenance**. An optional **Mechanistic Research** tab is hidden or marked experimental until gate G7 passes. Every prediction view includes a plain-text target unit, measured-versus-predicted distinction, applicability status, model version and source experiment details.

7.3 Model selection and holdout state machine

```
UNVERIFIED_SOURCE
  -> SOURCE_VERIFIED
  -> DATA_AUDITED
  -> TARGET_APPROVED
  -> SPLIT_FROZEN
  -> DEVELOPMENT_TUNING
  -> CHAMPION_APPROVED
  -> FINAL_TEST_EVALUATED (once as the primary result)
  -> REPORT_APPROVED
  -> RESEARCH_DEMO_READY

Any failed gate -> BLOCKED(reason, remediation)
Any revised target/split/config after final test -> NEW_PROTOCOL_VERSION
```

The holdout lock shall use an approved manifest and an evaluation ledger entry identifying the model/config version. A second holdout run may be performed for diagnostic reproducibility, but is not evidence of fresh independent validation. Any change inspired by its errors must be called *post-hoc exploratory development* and must not replace the first unbiased holdout record.

Research no-go rules: lack of a usable target; unresolved critical units; unverifiable source integrity; no group-safe evaluation possible; severe time leakage; no reproducibility; or insufficient independent experimental data to support stated inference. A no-go outcome is a valid research result.

8. Scientific model and validation specification

8.1 Inference contract and sample construction

V1 shall estimate a **measured offline target at a measured offline time** using only inline sensors that would have been available by that cutoff. It shall not use prior offline targets in this primary sensor-only track. If a separate lab-assisted track is later designed, the most recent offline target is eligible only when its *report availability time*, not just collection time, precedes the cutoff; the track must be evaluated and reported separately.

Modeling samples shall be constructed by linking one real offline assay with one point-in-time feature snapshot, or an explicit unscorable/abstained status. There shall be no pseudo-sample inflation from interpolation of target values. Cross-sectional sample weights must not make long cultivations dominate independent-experiment generalization claims.

8.2 Split and cross-validation specification

Partition by experiment_id so both parallel reactors remain together. The default proposed final test is two experiment groups, ideally representing batch and fed-batch if coverage and scientific design permit. This is a **candidate protocol, not a predetermined assignment**; frozen IDs and stratification decisions must be recorded before model tuning. Development hyperparameters are selected only through experiment-group-aware folds. Since fed-batch has three experiments, confidence about cross-mode generalization will be limited even under a balanced holdout.

If adequate target coverage across independent groups is lacking, the team shall reassess the task and possibly prefer grouped leave-one-experiment-out evaluation with nested tuning and no exaggerated independent-test claim. The report must make clear which protocol was actually executed.

8.3 Model families and fair comparison

- **Naive:** training-group mean/median or a declared operationally available sensor-only proxy. A last-known offline target baseline is **not** an equal sensor-only baseline unless explicitly designated lab-assisted.
- **Transparent statistical:** calibrated single-sensor relationship; Ridge/Elastic Net; partial least squares regression (PLS) for correlated sensor families.
- **Nonlinear:** Random Forest or histogram-based gradient boosting, subject to group-count and feature-dimension constraints.
- **Optional uncertainty:** Gaussian process or group-aware prediction interval method only with documented assumptions and appropriate evaluation coverage.
- **P2 mechanistic:** transparent ordinary differential equations with units, initial states, parameter ranges, solver verification and identifiability analysis. No data-backed calibration is claimed when measured states/feeds do not support it.
- **P2 hybrid:** mechanistic baseline plus train-only statistical residual model, evaluated against both components under identical grouped folds.

Model families should receive comparable allowed inputs, scoring rows, fold assignments and preprocessing policies. When sample availability differs, show the difference instead of silently using disjoint scoring sets and comparing errors as if they were equivalent.

8.4 Metrics and reporting

The **proposed** primary ranking is macro-averaged experiment-level MAE in the original target unit. For each experiment compute MAE, RMSE, signed bias, valid scored assay count, abstention count and baseline-relative delta. Secondary summaries include median/worst-group error and process-mode stratification. R-squared is optional and must be interpreted carefully for low-variance per-run targets.

Do not invent a scientific acceptance threshold. After the target and units are established, solicit scientific review to define meaningful error tolerance. A model may be declared **best among compared methods**, not automatically **fit for operational use**.

Uncertainty estimates must be marked as calibrated or uncalibrated, with empirical coverage and interval width where estimable. Repeated timepoints and paired reactors cannot be treated as fully independent observations for confidence intervals. If calibration data are too limited, publish descriptive variability and abstain from coverage guarantees.

8.5 Explainability, failure and robustness

Evaluate at least one sensor-family ablation (e.g., turbidity/permittivity versus gas signals when corresponding fields exist), a missing-sensor stress test, a batch/fed-batch comparison, and a per-experiment error table. Feature attribution should use approaches appropriate for correlated signals and should not claim causal biochemical relevance. An explanation is a description of model behavior, not a demonstrated biological mechanism.

Preserve negative results, instability, out-of-distribution warnings and nonconverged candidate runs. A candidate that fails on fed-batch experiments cannot be called broadly generalizable simply because overall pooled RMSE appears low.

8.6 Optional mechanistic feasibility gate

The mechanistic extension shall not claim a calibrated biological state model unless data audit confirms required concentrations, volumes/feed schedules, known units, time origins, plausible assumptions and sufficient independent evidence for identifiability. Sensitivity/parameter-correlation or multi-start analyses are mandatory before interpreting fitted parameters. Otherwise its solver may remain a synthetic example outside the main empirical evaluation, clearly described as illustrative.

9. Verification, QA and security test plan

9.1 Automated unit/integration tests

Test ID	Fixture / stimulus	Expected result	Coverage
T-001	One byte changed in publisher ZIP	Checksum hard-fail and no canonical output	DAT-001, NFR-003
T-002	Missing/duplicate reactor file	Reconciliation failure with file and ID details	DAT-003
T-003	Unknown units, malformed timestamp, nonnumeric value	Explicit quality/error record; raw preserved	DAT-004/006, NFR-005
T-004	Two reactors with same experiment assigned to different partitions	Split validation hard-fail	FEA-006/007, NFR-004

Test ID	Fixture / stimulus	Expected result	Coverage
T-005	Future sensor values perturbed after time t	Earlier feature snapshots and predictions unchanged	FEA-003/004, NFR-004
T-006	Interpolated offline pseudo-label inserted	Scoring rejects pseudo-label; assay count unchanged	FEA-002
T-007	Test-group data leaked into scaler fitting	Fitted-group audit detects and fails run	MOD-003
T-008	Unequal experiment sizes with known error values	Macro group MAE weights experiments equally	MOD-005/006
T-009	Tune candidate using final holdout before approval	Access refused and attempt logged	MOD-007
T-010	Missing mandatory sensor at inference	abstained or unsupported; no fabricated point estimate	FEA-008, UI-004
T-011	Interval coverage outside calibration range	Uncalibrated/unsupported status shown, not a reliability claim	MOD-009
T-012	Model worse than baseline in one experiment	Negative result remains in model comparison/report	MOD-011, NFR-006
T-013	Dashboard launched on read-only artifact folder	All review views work; no writes or fit calls	UI-007, NFR-012
T-014	Corrupt or untrusted model artifact	Loading fails securely without code execution	NFR-011
T-015	Re-run full workflow with pinned environment	Deterministic partitions and numerical parity	NFR-001/002
T-016	License unresolved, public package requested	Packaging omits raw data and includes citation/legal warning	NFR-010
T-017	Mechanistic state units missing	Data-fitted ODE/“hybrid validated” path blocked	MH-001/002
T-018	Scenario values outside declared bounds	Scenario rejected or flagged unsupported	MH-005

The test fixtures shall be small and **synthetic**, designed specifically to trigger failure conditions. They shall not be recycled as evidence of predictive performance on real CHO cultivations.

9.2 Acceptance and reproducibility protocol

An independent reviewer receives a pinned-environment setup command, source DOI/download instructions, expected checksum, experiment manifest, approved target config and frozen split, plus a minimal command sequence to recreate canonical features, all accepted model metrics and the research dashboard. Exact floating-point equality is not required if the solver/BLAS backend differs; tolerances must be defined and justified, not silently widened after failure.

Reviewers must verify traceability from at least one visible prediction back to model run, cutoff feature snapshot, reactor ID, assay reference and source digest. They should also observe one failed/unsupported prediction and at least one case where a more complex model does not outperform its baseline.

9.3 Security and public-portfolio review

Never load arbitrary user-provided pickle/joblib files or execute uploaded scripts. Ensure run manifests reference trusted local paths only; resolve paths against an allowed artifact root and reject traversal or symlinks escaping that root. Package no secrets, credentials, private data or unlicensed raw dataset. Generate a software dependency and license inventory before public release. Keep BioSynvia as a working title until similarity review; a different spelling is not equivalent to trademark clearance.

10. PRD-to-SRD traceability matrix

This section provides requirement coverage against the **unchanged** BioSynvia PRD v1.1. SR-* IDs identify proposed system controls; T-* IDs are acceptance tests from §9. No implementation status is asserted.

PRD ID	Relevant SRD requirements	Primary verification
FR-01	DAT-001, DAT-002, NFR-003	T-001
FR-02	DAT-003	T-002
FR-03	DAT-004, DAT-006, FEA-001	T-003
FR-04	DAT-005, FEA-002	T-005, T-006
FR-05	DAT-007	Target contract/approval fixture
FR-06	FEA-006, FEA-007	T-004
FR-07	FEA-003, FEA-004, FEA-005	T-005
FR-08	MOD-001, MOD-002, MOD-003	Group-CV baseline comparison, T-007
FR-09	MOD-004, MOD-007	T-009, T-015
FR-10	MOD-005, MOD-006, MOD-011	T-008, T-012
FR-11	UI-001, UI-002, UI-003, UI-004	T-013 and review demonstration
FR-12	UI-005, NFR-010	Research-only banner and packaging review
FR-13	FEA-008, MOD-009, UI-004	T-010, T-011
FR-14	MOD-008	Sensor ablation and mode-report checks
FR-15	MOD-010, UI-006, NFR-002	T-015 and lineage traversal
FR-16	MH-001, MH-002, MH-003	T-017 and solver test suite
FR-17	MH-004	Same-split paired comparator regression
FR-18	MH-005	T-018
NFR-01	NFR-001	T-015

PRD ID	Relevant SRD requirements	Primary verification
NFR-02	NFR-002	Lineage traversal
NFR-03	DAT-008, NFR-003	T-001 and idempotency
NFR-04	FEA-002/003/006, NFR-004	T-004/005/006
NFR-05	DAT-006, FEA-008, NFR-005	T-003/010
NFR-06	MOD-011, NFR-006	T-012
NFR-07	NFR-007	Offline/privacy review
NFR-08	UI-008, NFR-008	Accessibility inspection
NFR-09	NFR-009	Local-only execution audit
NFR-10	NFR-010	T-016
NFR-11	NFR-011	T-014

11. Build phases, dependencies, deliverables and exit gates

Phase	Dependencies	Estimated effort	Required deliverables	Exit decision
P0 — Research setup	PRD v1.1	3–5 hours	Project charter, no-cloud baseline, repo, pinned environment, source plan	Begin dataset audit
P1 — Data intake/audit	P0, source access	8–12 hours	Source manifest, inventory, unit dictionary, assay coverage, raw/aligned comparison	**G1:** usable measured target and provenance?
P2 — Canonical + features	G1, scientific target decision	8–12 hours	Versioned canonical tables, group manifests, leakage-safe snapshots and tests	**G2–G3:** approved splits and valid features?
P3 — Model bench	G3	9–14 hours	Baselines, PLS/Ridge, nonlinear comparator, grouped CV, run registry	**G4:** benchmark protocol complete?
P4 — Scientific validation	G4, frozen champion	11–18 hours	First unbiased holdout, per-run failure analysis, robustness, model card	**G5:** honest, reproducible evidence?
P5 — Demo and report	G5	10–16 hours	Read-only dashboard, technical presentation, repository docs, QA report	**G6:** reproducible research demo?
P6 — Advanced extension	G7 + scientific signoff	18–35 additional hours	ODE feasibility analysis, solver validation,	Optional; not required for credible V1

Phase	Dependencies	Estimated effort	Required deliverables	Exit decision
			hybrid comparison or synthetic appendix	

All estimates are planning ranges, not deadlines or promises of achieved accuracy. Work pauses at a failed scientific exit gate; the appropriate response is to change target/scope or report no-go, not to fabricate a successful model.

11.1 Priority implementation backlog

Order	Task	Output	Acceptance test
1	Verify source archive and rights	source_manifest.json	T-001, T-016
2	Inventory original and aligned source files	dataset_inventory.csv	T-002
3	Audit timestamps, target candidates, units and missingness	field_dictionary.csv, analysis report	T-003
4	Approve target/inference contract	target_decision.yaml, inference_contract.md	DAT-007 approval gate
5	Freeze experiment groups	splits_v1.json	T-004
6	Build snapshot generator	feature_snapshots.parquet	T-005/006
7	Fit baselines and statistical models	model runs, CV metrics	T-007/008
8	Select champion and evaluate locked holdout	first-test ledger, model card	T-009/012
9	Implement read-only review app	Streamlit workbench	T-010/013
10	Reproduce/inspect and prepare interview report	signed QA + concise presentation	T-015 and independent review

11.2 Milestone acceptance criteria

- **M1 Data-ready:** DOI/checksum, experiment/reactor structure, target and units verified; documented go/no-go.
- **M2 Model-ready:** group splits frozen, sensor-only inference contract, future-perturbation and preprocessing isolation tests pass, valid baseline established.
- **M3 Research-ready:** immutable first holdout run, per-experiment metric table, failure analysis, limitations and model card; clean reproduction passes.
- **M4 Demo-ready:** UI shows measured/predicted/missing states and traceable model version; no production/GxP/CGT claims; publication gates reviewed.
- **M5 Advanced-ready (optional):** mechanistic states/units identifiable or synthetic-only; numerical solver tests and same-split comparisons completed.

12. Risks, decisions and controlled changes

Risk ID	Risk	Severity	Trigger	Required response
R-01	Name confusion with SYNVIA	High	Commercial/public launch or brand claim	Keep BioSynvia provisional; obtain authoritative clearance or change name
R-02	Unknown dataset redistribution license	High	Public repo/package includes source rows	Exclude raw files until permissions confirmed
R-03	Target lacks adequate assay coverage	High	Phase 1 target audit below viable independent-group count	Choose defensible alternative or no-go
R-04	Paired-run experiment leakage	Critical	Same experiment appears in development and test	Block and invalidate affected evaluation
R-05	Future-data contamination	Critical	A future sensor or offline target changes earlier features	Block and rebuild all affected features/models
R-06	Small-sample/paired dependence	High	Unstable folds and high group variance	Report spread, adjust protocol; no broad claims
R-07	Mechanistic parameter unidentifiability	High	Multiple indistinguishable fits, missing units/feed details	Restrict to synthetic research and disclose
R-08	Product-style dashboard overstates maturity	High	Model outputs labeled production/reliable without evidence	Reject UI/review release
R-09	Unapproved cloud charges or CI runs	High	Hosted services/workflows activated	Disable before deployment; manual local-only testing
R-10	Security exposure through model serialization	High	Untrusted pickled artifact loaded	Trusted-local-only loading, validate paths/hashes

Change control: Any modification to target definition, input feature set, experiment partition, label construction, test interpretation or accepted model claim shall increment a protocol version, run the affected tests and update the model card and traceability. A change informed by final-test results cannot quietly supersede the originally frozen holdout evaluation.

12.1 Open decisions and accountable reviewers

Decision ID	Decision required	Responsible reviewer	Latest permissible point
D-01	Is a viable measured offline target available, with known	Developer + scientific reviewer	Before P2

Decision ID	Decision required	Responsible reviewer	Latest permissible point
	units and sufficient independent groups?		
D-02	What is the exact sensor-only inference-time feature eligibility rule?	Developer + data reviewer	Before features
D-03	Which experiment groups are assigned to the final test, and why?	Developer + methodology reviewer	Before tuning
D-04	Is the publisher's raw dataset redistributable?	Owner/rights holder	Before public repo
D-05	What error magnitude would be scientifically meaningful for the selected target?	Scientific reviewer	Before any use-utility claim
D-06	Are mechanistic states, feed and volume inputs sufficient for data-backed ODE calibration?	Scientist + modeler	Before P6
D-07	Is BioSynvia safe for public commercial branding given SYNVIA similarity?	Brand owner/legal reviewer	Before public launch

13. Final system acceptance and evidence package

Research acceptance is not manufacturing acceptance. BioSynvia V1 may be described as a *reproducible CHO bioprocess soft-sensing research workbench* only when all required gates have evidence; it must not be described as a validated manufacturing Digital Twin.

Exit check	Evidence required	V1 classification
Data verified	Publisher checksum, data dictionary, paired experiment inventory	Mandatory
Target credible	Measured target, unit, coverage and protocol approval	Mandatory
Time/group integrity	Group-split, future perturbation and measured-only scoring tests	Mandatory
Fair baselines	Naive + transparent + nonlinear candidates under identical folds	Mandatory if feasible; exceptions recorded
Honest model evidence	First holdout, per-experiment metrics, negative outcomes, failure analysis	Mandatory
Traceability	Source-to-prediction-to-report lineage reproducible	Mandatory
Demonstration	Local read-only workbench with units, missingness and limitations	Mandatory

Exit check	Evidence required	V1 classification
Rights and naming	Raw distribution withheld if rights unknown; BioSynvia provisional mark	Mandatory for public portfolio
Mechanistic/hybrid extension	Scientific eligibility/signoff and comparator results	Optional, not required for V1
Production qualification	GxP validation, deployed control interface, commercial trademark approval	**Explicitly excluded from V1**

The demo-ready release should include: a local setup README, source DOI and checksum verification instructions, source/schema manifests, field and quality dictionaries, approved target/split protocol, reproducible model training and evaluation commands, CV and first-held-out metric tables, model card, failed-experiment discussion, read-only dashboard and 6-slide scientific case study.

Interview/portfolio language permitted only after reproduction: "Developed a research prototype for CHO cell-culture soft sensing; compared baseline, linear/chemometric and nonlinear models using experiment-group-held-out validation, documented uncertainty and failure cases, and created a traceable scientist-facing review interface." Specific accomplishments, technologies, metrics and improvements must match actual observed outputs; do not add the prototype as completed work before implementation.

14. Source register and document precedence

The source-of-truth hierarchy is: (1) actual rights-cleared publisher dataset and verified metadata for empirical facts, (2) **BioSynvia PRD v1.1** for product scope, research intent and decision gates, (3) this SRD for proposed implementation contracts, (4) later approved implementation designs and tested evidence for actual system behavior. Nothing in this SRD overwrites the existing PRD.

- **[R1] Primary dataset:** Uhlendorff, S., Fulek, R., Eimler, J., Pein-Hackelbusch, M. and Frahm, B. (2026). *Dataset Based on Chinese Hamster Ovary (CHO) Cultivations including Turbidity, Permittivity, O2 and CO2 Measurements*. DOI: 10.5281/zenodo.20829178. Dataset release record indicates 24 cultivations, 12 paired experiments, 38 inline and 10 offline variables; field-level contents require ingestion audit.
- **[R2] Product specification:** *BioSynvia PRD v1.1 — Problem Statement First*, November 4, 2026. Available unchanged as the user-approved product source.
- **[R3] Scientific methodology:** scikit-learn LeaveOneGroupOut documentation and PRD §7 experiment-design rules.
- **[R4] Research and legal context:** USPTO Likelihood of confusion guidance; the scientific-software-related SYNVIA name conflict remains unresolved.
- **[R5] Role alignment context:** User-provided July 2026 J&J Senior Scientist/Modeler job description, used for portfolio relevance only; no claim of employment, sponsorship approval, or still-open requisition.

Specification status: REVIEW DRAFT, NOT IMPLEMENTED. The working name stays **BioSynvia** exactly. Creation of this standalone SRD does not edit, rename, or replace the PRD.